

ARUL GIRISH JAGANATHAN

STAFF EMBEDDED SOFTWARE ENGINEER

Bengaluru, India | tech.arulgirish@gmail.com | +91-7010041408 | LinkedIn

PROFESSIONAL SUMMARY

Technology Lead and embedded software engineer with nearly 9 years building firmware and real-time software for medical devices, automotive ECUs, and industrial systems. Strong in Embedded C/C++, Python, ARM Cortex-M, RTOS, bare-metal firmware, hardware/software integration, and low-level debugging. Owned firmware across FDA Class II/III medical devices and ISO 26262 ASIL-D automotive systems, with experience in firmware architecture, state machines, communication protocols, power management, system bring-up, validation, and safety-critical engineering.

LEADERSHIP & SCOPE

- Owned real-time firmware and product-level features, taking work from technical feasibility and design decisions through implementation, validation, and release.
- Led firmware ownership across three connected medical-device components and worked across firmware, hardware, system, QA, product, software, and verification teams.
- Solved field and integration problems at system level, including sensor bring-up under hardware delay, sealed-device recovery, and Tier-1 automotive crash-algorithm issues.

CORE SYSTEMS & TECHNICAL DEPTH

Silicon & architectures: ARM Cortex-M/A; STMicroelectronics STM32; Microchip SAM4S; Panasonic PAN1326; Broadcom BCM2837; PIC MCUs.

Languages & toolchains: Embedded C; C++ (bare-metal & RTOS); STL; Python; VBA (Automation Workbooks); GCC; MATLAB / Simulink / Target-Link (M-Script Automation & Code Generation).

OS & runtimes: FreeRTOS; Embedded Linux; bare-metal.

Core systems: Multi-threads; IPC; state machines; memory optimization; fault handling; task scheduling; DMA.

Power & hardware: Dual-Rail Power Budgeting; PMIC Sequencing; Dynamic Clock Gating; Multi-Channel ADC Sampling; Hardware PWM Loops; Qi Wireless Power Modulation.

Diagnostics & debugging: Logic Analyzers; Mixed-Signal Oscilloscopes; JTAG/SWD Debuggers (Segger J-Link); GDB; Valgrind Memory Leak Profiling; Dynamic Run-Time Analysis.

Connectivity & protocols: BLE 5.x; Medical-USB Host/Device; IrDA Stacks; CAN Bus; I2C; SPI; UART/USART; Modbus RTU/TCP; Series50.

Safety & validation: IEC 62304 (Class C); ISO 14971; ISO 26262 (ASIL-D); FMEA; CPPuTest; VectorCast; Coverity; Bullseye Branch Coverage; TiCS; GitLab; SVN; Jenkins; MDSng; Crash-Bench; IBM DOORS.

PROFESSIONAL EXPERIENCE

TECHNOLOGY LEAD

Dover Corporation | Bengaluru, India | Nov 2025 – Present

Systems Platform: Markem-Imaje Industrial Printing Platforms and Print & Apply 2200 Series

- Real-Time Firmware:** Owned the real-time firmware feature engine for industrial printing, covering motor control and printhead synchronization. Improved task scheduling and synchronization to keep print cycles deterministic at production speeds, increasing throughput from 60 to 65 labels/minute across 1,000+ production lines. The additional 5 labels/minute increased sustained production capacity on every affected line, accelerating product flow into downstream operations across the installed base.
- Workflow Automation:** Mapped manual build/test steps and developed Python and Bash automation to combine code-translation steps and automate initial hardware checks; reduced manual work and improved engineering throughput by 2x, helping the team release features faster.
- Technical Leadership:** Led feature planning and technical decision-making across firmware, hardware, QA, system, product, and software teams; assessed feasibility, planned releases, and took changes through implementation and validation while helping improve defect-resolution during critical releases.

LEAD DEVELOPER

GE Healthcare | Bengaluru, India | Mar 2023 – Oct 2025

Systems Platform: MIC CoroCare Connected Fetal/Maternal Wearable Monitoring Ecosystem (FDA Class II/III)

- Novii+ Wearable Pod:** Owned the main firmware application layer on Microchip SAM4S (ARM Cortex-M4), Bare-Metal C, Panasonic PAN1326 BLE, IrDA, Series50, I2C, multi-channel ADC, PMIC, and NXP FXLS8974CFR3 accelerometer; handled real-time maternal/fetal heart rate, fetal movement, uterine activity, and other patient data, including Bluetooth/IrDA packet handling, synchronization, and power-conscious transfer.
- Sensor Migration & Validation:** When the legacy wake-up accelerometer was discontinued, worked around a three-month sensor-PCB bring-up delay by connecting I2C probe lines from existing pod hardware to a TI development kit; debugged signal attenuation and EMI, stabilized I2C, and kept firmware development and validation moving before production boards were available.
- Novii+ Charger / SPC:** Owned charger development from design through testing and release on STM32L0xx (ARM Cortex-M0+), BLE, IrDA, Series50, Qi Wireless Charging, Hall sensors, I2C, GPIO/PWM, and Keil; designed the state machine for startup, idle, pairing, active, low-power, undocking, and fault handling and integrated Hall sensing for pod placement, Qi charging, and LED status.

- **System Recovery & Diagnostics:** For a sealed Pod with no hard reset or OTA recovery, reused the existing two-minute idle-timeout shutdown and programmed the SPC to cut Qi power for exactly 150 seconds, leaving a 30-second buffer for timeout, power-down, and reset; charging then resumed and the Pod booted ready for pairing. Also implemented foreign-object, thermal, low-power, and calibration fault detection, logging, and recovery.
- **Novii+ Adapter / NIM:** Led firmware on STM32L0xx connecting the wearable to the CoroCare monitor through Medical-USB; designed the Bluetooth/IrDA interface state machine, authentication, parsing, and transfer paths, and added runtime checks for patch status and wireless issues, packet-loss recovery, and isolation of pod-side faults.

SENIOR SOFTWARE ENGINEER

Robert Bosch | Bengaluru, India | Jan 2020 – Mar 2023

Systems Platform: Automotive Airbag Electronic Control Units (ECUs) — ISO 26262 / ASIL-D | Embedded C/C++, RTOS, MATLAB, Simulink, Target-Link, Microsoft Visual Studio, VectorCast, MDSng, Crash-Bench, IBM DOORS

- **Crash Algorithms & Production Integration:** Developed and verified crash-deployment algorithms in an ASIL-D environment using Embedded C/C++, RTOS, MATLAB/Simulink, and Target-Link; integrated optimized hand-written C into production ECU software and supported calibration for the Tata Punch program, which achieved a 5-Star Global NCAP safety rating. Converted finalized code into DLLs with Microsoft Visual Studio for system integration.
- **Signal Processing & Simulation:** Analyzed raw multi-axis crash sensor data with Crash-Bench, processed and normalized data for Bosch calibration requirements, integrated generated DLLs into MDSng, calibrated sensor-trigger conditions, and verified deployment timing against RTTF requirements.
- **Validation & Traceability:** Created module-level test plans and performed unit testing in VectorCast; maintained requirement-to-code traceability in IBM DOORS and supported ASIL-D verification.
- **Customer Field Claims:** Worked directly with Tier-1 automotive customers to clarify requirements and provide technical solutions; used telemetry and crash data to isolate algorithm issues and work through software fixes for specific crash scenarios.

EMBEDDED DEVELOPER

3Q Technologies | Coimbatore, India | Jan 2018 – Jan 2020

Systems Platform: Industrial IoT Smart SCADA Solar Controllers

- **Industrial IoT Gateway:** Developed a Python telemetry gateway on Embedded Linux/Raspberry Pi 3 Model B+ (Broadcom BCM2837 ARM Cortex-A53), integrating serial communication with solar inverters and Modbus RTU/TCP; handled around 5,000 inverter nodes and optimized throughput to roughly 50,000 telemetry data points/minute.
- **Cloud & Deployment:** Built the pipeline to remote MySQL for web/Android monitoring; worked directly with customers during deployments, collected site requirements, assessed technical constraints, and supported implementation at customer locations. Used MySQL, phpMyAdmin, and Web/Android API endpoints.

EDUCATION

Bachelors in Electronics and Communication Engineering | Anna University – SREC